

Production MLP Multi-Seed Training Fixed-Table Evaluation Against SVGP

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Claim: the production MLP does *not* beat the Stage 3 SVGP on the cleanest point-level CDF metric, but it gets close while producing much more conservative uncertainty intervals.

- Main uncensored CDF RMSE: SVGP 4.193 mm vs. production MLP 4.265 ± 0.040 mm.
- Production MLP has higher coverage: $1\sigma = 0.887$, $2\sigma = 0.989$, vs. SVGP 0.716/0.945.
- The diagnostic `mu_anchor + raw_uncertain` MLP is worse on point-level CDF, but gives the best Q1 extrapolation RMSE.

Thesis framing

SVGP should be presented as a strong probabilistic baseline, not a strawman. The MLP result is competitive in mean accuracy, more conservative in uncertainty, and easier to deploy; the μ -anchor path is useful mainly as an extrapolation diagnostic.

Compared Model Families

Model family	Run / seed contract	Role in comparison
SVGP Stage 3	seed 42; two-stage heteroscedastic sparse GP	Strong probabilistic baseline trained with Stage 3 data: synthetic Q1 plus real clean CDF.
Production MLP	5 seeds: 7, 17, 42, 99, 2024; Stage 3 anchor_off	Production candidate. Warm-started from full-data Stage 2 no_anchor; trained with production KD loss.
μ-anchor raw-uncertain MLP	5 seeds: 7, 17, 42, 99, 2024; $\lambda_{\text{anchor}} = 0.01$	Diagnostic candidate. Warm-started from Stage 2 mu_anchor; uses uncertain-bin raw/KD blend.

All neural models use the same engineered feature family: time_norm, tilt, plumes, injection duration, backpressure, $\log p_{\text{inj}}$, and $\log p_{\text{chamber}}$.

Production MLP Training Recipe

Architecture and input

- MLP hidden widths: 512, 512, 128
- Dropout: 0.30
- Output: Gaussian mean and log variance
- Target scale: engineered A -scaled penetration
- Stage 3 time grid: 512 points over 0–5 ms

Training schedule

- Base Stage 2 teacher: no_anchor
- Stage 3 refinement: 150 epochs
- Batch size: 128
- Weight decay: 10^{-4}
- Loss: raw CDF NLL + production KD
- KD mode: mse_mu_plus_sigma
- KD σ weight: 5.0
- Stage 3 anchor: off, $\lambda_{\text{anchor}} = 0$

Production run roots: MLP/runs_mlp/distill_cdf_onset_v2_ablate_anchor_off_20260521_122435 through ..._124004. The five seeds are evaluated independently and reported as mean $\pm$ std.

Training Data Contract

Training source	train pts	val pts	test pts	Use
Synthetic Q1 / Stage 3 base	645,120	138,240	138,240	Shared synthetic supervision
Real clean CDF, row split	589,741	103,201	–	Raw CDF refinement
Combined SVGP Stage 3	1,234,861	241,441	138,240	SVGP seed-42 training table

Fairness note

The production MLP and SVGP are compared after both consume the same full-data Stage 3 information class: synthetic Q1 supervision plus real clean CDF trajectories. LONO is kept separate because it answers a different question: held-out nozzle-family transfer.

SVGP source: `seed_config.json` in the seed-42 Stage 3 GP run folder.

Evaluation Protocols

Protocol	points	conditions	Purpose
Full clean CDF	692,942	592	Legacy full-series metric; includes right-censored regions.
CDF uncensored	542,565	592	Main physical point metric after conservatively removing FOV and signal-decay censoring.
P50 observed	4,213	495	Condition-level P50 points used by the Q1 oracle fit.
Q1 grid, extrapolated	21,527	495	Robust extrapolation check outside the observed P50 window.

The Q1 protocol is indexed by `MLP/synthetic_data/p50_q1_oracle/p50_q1_fit_metrics.csv`. Model errors are computed on the companion observed/grid tables sharing the same `condition_id`: `p50_q1_observed_fit_points.csv` and `p50_q1_predictions.csv`.

Full Clean CDF: Legacy Reference

Model	RMSE mm	MAE mm	Bias mm	P95 mm	cov1	cov2
SVGP Stage 3	4.628	3.173	-0.070	9.242	0.703	0.939
Production MLP, mean(5)	4.735 $\pm$ 0.037	3.293 $\pm$ 0.039	+0.626	9.167	0.867	0.985
μ -anchor raw-uncertain, mean(5)	5.200 $\pm$ 0.048	3.654 $\pm$ 0.047	-0.585	10.525	0.834	0.985

- This is the earlier full-clean metric with right-censored regions retained.
- SVGP wins pointwise RMSE/MAE; production MLP wins coverage and slightly lower P95.
- The μ -anchor diagnostic model is not a production winner on this protocol.

Main Result: CDF Uncensored

Model	RMSE mm	MAE mm	Bias mm	P95 mm	cov1	cov2
SVGP Stage 3	4.193	2.894	-0.040	8.461	0.716	0.945
Production MLP, mean(5)	4.265 $\pm$ 0.040	3.032	+0.562	8.307	0.887	0.989
μ -anchor raw-uncertain, mean(5)	4.941 $\pm$ 0.066	3.505	-0.868	10.123	0.840	0.988

Reading

On the cleanest point-level metric, production MLP is close but still behind SVGP: +0.073 mm RMSE. The main compensating advantage is conservative uncertainty, not lower mean error.

P50 Observed: Condition-Level P50 Check

Model	RMSE mm	MAE mm	Bias mm	P95 cond. mm	RMSE mm
Q1 oracle fit	1.010	0.689	+0.197	2.048	0.900
SVGP Stage 3	2.649	1.603	+0.358	5.900	1.855
Production MLP, mean(5)	2.848 ± 0.059	1.826	+0.856	6.017	2.068
μ -anchor raw-uncertain, mean(5)	3.212 ± 0.245	2.147	-0.468	7.323	2.837

- Q1 oracle is a reference fit to P50 points, not a learned model competing for deployment.
- Among learned models, SVGP remains the best on observed P50 points.
- The μ -anchor model does not improve the observed-window mean prediction.

Q1 Extrapolation: Where the μ -Head Helps

Model	RMSE mm	MAE mm	Bias mm	P95 cond. mm	RMSE mm
SVGP Stage 3	10.591	7.236	+6.410	23.701	7.465
Production MLP, mean(5)	10.864 $\pm$ 0.064	7.380	+6.753	24.514	7.461
μ -anchor raw-uncertain, mean(5)	10.358 $\pm$ 0.106	6.965	+6.269	23.768	7.072

Interpretation

The μ -anchor/raw-uncertain recipe hurts local pointwise CDF accuracy but reduces the Q1 extrapolation error and positive bias. This supports using it as a diagnostic for stable extrapolation, not as the main production model.

What the Comparison Supports

- 1 **SVGP is the accuracy winner** on the two most direct observed-data metrics: full clean CDF and uncensored CDF.
- 2 **Production MLP is close enough to be credible**: on uncensored CDF it is only 0.073 mm RMSE worse than SVGP, while maintaining much higher empirical coverage.
- 3 **The μ -anchor head has a different behavior**: worse observed-data RMSE, better Q1 extrapolation RMSE and lower positive extrapolation bias.
- 4 **The thesis should separate claims**: use SVGP as the strong probabilistic baseline; use production MLP for deployability and conservative uncertainty; use μ -anchor/raw-uncertain as evidence about extrapolation regularization.

Evaluation script and reports

- `MLP/eval/evaluate_stage3_fixed_tables.py`
- `comparison_reports/stage3_fixed_table_eval_20260521/main_table.md`
- `comparison_reports/stage3_fixed_table_eval_20260521/group_summary.csv`
- `comparison_reports/mlp_production_stage3_vs_svgp_stage3_full_clean_20260521/`

Main model roots

- SVGP: `gp_baseline_stage3_20260521_112229/per_seed/seed_42/model.pt`
- Production MLP: `distill_cdf_onset_v2_ablate_anchor_off_20260521_122435 ... 124004`
- μ -anchor diagnostic:
`distill_cdf_onset_v2_mu_anchor_raw_uncertain_raw05_kd025_prod_kd_20260521_124745 ... 125816`

This deck:

`Thesis/slides/slides_production_mlp_svgp_eval/slides_production_mlp_svgp_eval.tex`